



AIRFLOW COMPLETE CHEAT SHEET

1. Ubuntu basic setup

Update packages

```
sudo apt update
```

Meaning: Ubuntu ke available packages ki latest information update karta hai.

Install curl

```
sudo apt install -y curl
```

Meaning: curl install karta hai, internet se files/data download karne ke kaam aata hai.

2. Airflow project folder

Main Airflow folder banao

```
mkdir -p ~/airflow-class
```

Meaning: airflow-class naam ka folder banata hai.

DAG folder banao

```
mkdir -p ~/airflow-class/dags
```

Meaning: Is folder ke andar Airflow ki Python DAG files rakhenge.

Folder me jao

```
cd ~/airflow-class
```

Meaning: Airflow project directory me enter karta hai.

Current location check

```
pwd
```

Meaning: Abhi tum kis folder me ho, uska full path dikhata hai.

Expected:

```
/home/saqibkhan1691/airflow-class
```

Files/folders dekho

```
ls
```

Meaning: Current folder ki files/folders dikhata hai.

Detailed:

```
ls -l
```

Meaning: Files ke saath permissions, size, date etc. dikhata hai.

3. Python virtual environment

Virtual environment activate

```
source .venv/bin/activate
```

Meaning: Airflow ka Python environment activate karta hai.

Prompt me:

```
(airflow-class)
```

dikhna chahiye.

Airflow Home set karo

```
export AIRFLOW_HOME=$HOME/airflow-class
```

Meaning: Airflow ko batata hai ki uska main data/config/database folder airflow-class hai.

Check karo

```
echo $AIRFLOW_HOME
```

Expected:

```
/home/saqibkhan1691/airflow-class
```

4. Airflow database

Database initialize/migrate

```
airflow db migrate
```

Meaning: Airflow ki database tables create/update karta hai.

Database check

```
airflow db check
```

Meaning: Check karta hai database properly accessible hai ya nahi.

5. Airflow start karna

Airflow standalone

```
airflow standalone
```

Meaning: Airflow ke required components ko ek saath start karta hai.

Isme normally:

```
API Server  
Scheduler  
DAG Processor  
Triggerer
```

start hote hain.

⚠ **Ye terminal running rehna chahiye.**

Stop:

```
Ctrl + C
```

6. Airflow web UI

Browser me:

```
http://localhost:8080
```

Meaning: Airflow ka web interface.

Admin password dekhna

```
cat ~/airflow-class/simple_auth_manager_passwords.json.generated
```

Meaning: Standalone ka generated admin password dikhata hai.

7. New terminal me Airflow use karna

Agar airflow standalone ek terminal me chal raha hai, **new terminal** kholo.

Then:

```
cd ~/airflow-class
source .venv/bin/activate
export AIRFLOW_HOME=$HOME/airflow-class
```

8. DAG Python file banana

New folder

Example:

```
mkdir -p ~/airflow-class/dags/airflow-practice-Saqib-Khan
```

Meaning: Assignment ke liye DAG ka folder banata hai.

Python DAG file create/edit

```
nano ~/airflow-class/dags/airflow-practice-Saqib-Khan/practice_dag.py
```

Meaning: Nano editor me practice_dag.py file open/create karta hai.

9. Nano ke important commands

Save

Ctrl + O

Meaning: File save/write karta hai.

Then:

Enter

Exit

Ctrl + X

Meaning: Nano se bahar aata hai.

Save + exit

Ctrl + O

Enter

Ctrl + X

10. DAG file check

DAG folder dekho

```
ls ~/airflow-class/dags
```

Meaning: DAG folders/files dekhta hai.

Specific folder:

```
ls ~/airflow-class/dags/airflow-practice-Saqib-Khan
```

Meaning: Check karta hai `practice_dag.py` present hai ya nahi.

11. Airflow DAG folder check

```
airflow config get-value core dags_folder
```

Meaning: Airflow currently kis folder se DAGs read kar raha hai.

Expected:

```
/home/saqibkhan1691/airflow-class/dags
```

12. DAG list

```
airflow dags list
```

Meaning: Airflow me detected saare DAGs dikhata hai.

Specific DAG search:

```
airflow dags list | grep airflow_practice
```

Meaning: DAG list me `airflow_practice` naam search karta hai.

Example:

```
airflow dags list | grep practical2
```

13. DAG import error check

```
airflow dags list-import-errors
```

Meaning: Python DAG files me import/syntax related errors check karta hai.

Agar:

```
No data found
```

toh import error nahi hai.

14. DAG refresh/reserialize

```
airflow dags reserialize
```

Meaning: DAG files ko Airflow database ke saath re-process/reserialize karta hai.

Useful jab:

```
DAG file hai  
BUT  
DAG list me nahi aa rahi
```

15. DAG pause/unpause

Unpause

```
airflow dags unpause airflow_practice_saqib
```

Meaning: DAG ko active karta hai.

Pause

```
airflow dags pause airflow_practice_saqib
```

Meaning: DAG ko temporarily stop/pause karta hai.

16. DAG manually run/trigger

```
airflow dags trigger airflow_practice_saqib
```

Meaning: DAG ka ek new manual run start karta hai.

17. DAG runs check

```
airflow dags list-runs airflow_practice_saqib
```

Meaning: Is DAG ke saare runs dikhata hai.

Successful runs only:

```
airflow dags list-runs airflow_practice_saqib --state success
```

Meaning: Sirf successful runs dikhata hai.

18. DAG ke tasks dekho

```
airflow tasks list airflow_practice_saqib
```

Meaning: DAG ke andar available tasks dikhata hai.

Tumhare assignment me:

```
check_system  
process_data  
generate_report
```

19. Individual task test

```
airflow tasks test airflow_practice_saqib check_system
```

Meaning: check_system task ko test karta hai.

Similarly:

```
airflow tasks test airflow_practice_saqib process_data
airflow tasks test airflow_practice_saqib generate_report
```

Meaning: Individual tasks ko test karta hai.

20. Task status of a DAG run

```
airflow tasks states-for-dag-run DAG_ID RUN_ID
```

Meaning: Kisi particular DAG run ke **saare tasks ka status** dikhata hai.

Example:

```
airflow tasks states-for-dag-run airflow_practice_saqib "RUN_ID"
```

21. DAG run ID dekhna

```
airflow dags list-runs airflow_practice_saqib
```

Output me:

```
run_id
```

milega.

Us `run_id` ko `states-for-dag-run` me use kar sakte ho.

22. Airflow configuration

Full config

```
airflow config list
```

Meaning: Airflow ki configuration settings dikhata hai.

Specific setting

```
airflow config get-value core dags_folder
```

Meaning: Specific config value check karta hai.

23. Airflow providers

`airflow providers list`

Meaning: Installed Airflow providers/integrations dikhata hai.

24. Connections

Connections list

`airflow connections list`

Meaning: Airflow ke configured connections dikhata hai.

Specific connection

`airflow connections get CONNECTION_ID`

Meaning: Particular connection ki information dekhta hai.

25. Variables

Variables list

`airflow variables list`

Meaning: Airflow variables dekhta hai.

Variable get

`airflow variables get VARIABLE_NAME`

Meaning: Particular variable ki value dekhta hai.

Variable set

`airflow variables set VARIABLE_NAME VALUE`

Meaning: New variable/value save karta hai.

26. Airflow components manually

Normally `airflow standalone` ye sab start kar deta hai.

Scheduler

```
airflow scheduler
```

Meaning: DAG scheduling/tasks scheduling handle karta hai.

DAG Processor

```
airflow dag-processor
```

Meaning: DAG Python files ko scan/process karta hai.

API Server

```
airflow api-server
```

Meaning: Airflow API/web backend server start karta hai.

Triggerer

```
airflow triggerer
```

Meaning: Deferred/trigger-based tasks handle karta hai.

27. Help commands

Airflow overall help:

```
airflow --help
```

DAG help:

```
airflow dags --help
```

Task help:

```
airflow tasks --help
```

DAG runs help:

```
airflow dags list-runs --help
```

Task status help:

```
airflow tasks states-for-dag-run --help
```

Meaning: Tumhare installed Airflow version me exact syntax batata hai.

Ye bahut useful hai because Airflow 3.x me kuch
commands/options purane tutorials se different hain.

PRACTICAL — Complete Flow

Agar kal practical me zero se Airflow DAG banana ho, toh ye sequence yaad rakho:

1. Setup

```
sudo apt update
sudo apt install -y curl
mkdir -p ~/airflow-class/dags
cd ~/airflow-class
```

2. Environment

```
source .venv/bin/activate
export AIRFLOW_HOME=$HOME/airflow-class
```

3. Database

```
airflow db migrate
```

4. Start Airflow

```
airflow standalone
```

5. New terminal

```
cd ~/airflow-class
source .venv/bin/activate
export AIRFLOW_HOME=$HOME/airflow-class
```

6. New DAG folder

```
mkdir -p ~/airflow-class/dags/airflow-practice-Saqib-Khan
```

7. DAG file

```
nano ~/airflow-class/dags/airflow-practice-Saqib-Khan/practice_dag.py
```

Code paste →

```
Ctrl + O  
Enter  
Ctrl + X
```

8. Check DAG

```
airflow dags list | grep airflow_practice
```

9. Error check

```
airflow dags list-import-errors
```

10. If needed refresh

```
airflow dags reserialize
```

11. Unpause

```
airflow dags unpause airflow_practice_saqib
```

12. Run

```
airflow dags trigger airflow_practice_saqib
```

13. Check run

```
airflow dags list-runs airflow_practice_saqib
```

14. Check tasks

```
airflow tasks list airflow_practice_saqib
```

15. UI

```
http://localhost:8080
```

Then:

```
DAG  
↓  
Run  
↓  
Tasks  
↓  
Task  
↓  
Logs
```

🌀 10 Commands Jo Actually Yaad Rakhne Hain

Agar exam me **sab commands yaad nahi rehengi**, at least ye yaad rakh:

```
cd ~/airflow-class
```

→ Airflow folder me jao

```
source .venv/bin/activate
```

→ Python environment activate

```
export AIRFLOW_HOME=$HOME/airflow-class
```

→ Airflow ka home set

```
airflow standalone
```

→ Airflow start

```
nano ~/airflow-class/dags/my_dag.py
```

→ DAG Python file banao/edit karo

```
airflow dags list
```

→ DAGs dekho

```
airflow dags list-import-errors
```

→ DAG errors check

```
airflow dags unpause DAG_ID
```

→ DAG activate

```
airflow dags trigger DAG_ID
```

→ DAG manually run

```
airflow dags list-runs DAG_ID
```

→ DAG runs check

Bas ye flow dimaag me rakho:

CREATE FOLDER
↓
CREATE .PY FILE
↓
WRITE DAG
↓
SAVE (Ctrl+O)
↓
EXIT (Ctrl+X)
↓
DAG LIST
↓
IMPORT ERROR CHECK
↓
UNPAUSE
↓
TRIGGER
↓
CHECK RUN
↓
UI → TASKS → LOGS